

RAKSHANET

Predictive Cash-Out Hotspot Intelligence & Multi-Hop Mule Interdiction Platform

SMART INDIA HACKATHON (SIH) PROBLEM STATEMENT ALIGNMENT

Halting Illicit Fund Dissipation During the "Golden Window" (First 180–240 Minutes) of Cyber-Financial Fraud

Target Deployers: Indian Cyber Crime Coordination Centre (I4C), National Cybercrime Reporting Portal (NCRP), State Cyber Crime Police Wings, Core Banking Systems (CFCFRMS).


```
| RISK FUSION & EXPLAINABLE AI (XAI) |
| - Fused Risk Score:  $S_{\text{fused}} = 0.60 * S_{\text{graph}} + 0.40 * S_{\text{geo}}$  |
| - Game-Theoretic TreeSHAP: Local Feature Attributions ( $\phi_i$ ) translating into Court Verdicts |
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|                                     |
|                                     |
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| TACTICAL INTERDICTION & STATUTORY LEGAL FRAMEWORK |
| - Section 91 Cr.P.C. Inter-Bank Freezing           |
| - CFCFRMS Bank Lien Confirmation Gateways          |
| - Mobile Beat Patrol & PCR Van GPS Dispatch        |
| - Section 63 BSA / 65B IEA Evidence Dossiers       |
| - Section 457 Cr.P.C. Citizen Restitution Engine   |
| - Public 4-Stage Victim Recovery Transparency Portal |
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2. POLYGLOT DATABASE TIER & SCHEMAS

2.1 Polyglot Architecture Rationale

National cybercrime infrastructure requires handling orthogonal access patterns: relational transaction consistency, deep graph link traversals, and high-speed spatial searches. RakshaNet avoids monolithic database bottlenecks by decoupling workloads across three specialized engines:

Database Engine	Primary Workload	Data Models / Indexes	Performance Metric
PostgreSQL 16 + PostGIS 3.4	Relational Ground Truth, Spatial Queries, Audit Trails	B-Trees on UUIDs, GiST R-Trees on `GEOMETRY(Point, 4326)`	P99 < 15ms relational lookups; sub-10ms spatial radius search
Neo4j 5 Community	Multi-Hop Layering, Ring Discovery, GraphSAGE Embeddings	Account Nodes, Directed `TRANSFERRED` edges with timestamp/amount	Sub-50ms 2-hop traversals via indexed native pointer chasing
Redis 7 Alpine	In-Memory Cache & Pub/Sub Event Broker	Key-Value hot state, WebSocket alert queues	Sub-2ms publish/subscribe latency across active browser sessions

2.2 Relational Table Schemas (PostgreSQL / PostGIS)

The schema enforces strict data types, foreign key constraints, and spatial geometry:

Table Name	Key Attributes & Data Types	Constraints & Indexes	Operational Purpose
`accounts`	`id` (UUID), `account_number` (VARCHAR), `holder_name`, `bank_name`, `ifsc_code`, `balance` (NUMERIC), `risk_score` (FLOAT), `account_age_days` (INT)	PRIMARY KEY (`id`), UNIQUE (`account_number`), INDEX (`ifsc_code`)	Stores banking entities and core KYC indicators used by SHAP explainability.
`transactions`	`id` (UUID), `source_account_id`, `destination_account_id`, `amount` (NUMERIC), `channel` (VARCHAR: UPI/IMPS/ATM), `timestamp` (TIMESTAMPTZ)	FOREIGN KEY to `accounts.id`, INDEX (`timestamp`, `source_account_id`)	Double-entry ledger recording all high-velocity inter-bank transfers.
`complaints`	`id` (UUID), `acknowledgement_no` (VARCHAR 32), `loss_amount` (NUMERIC), `victim_account_id`, `suspect_account_id`, `location` (GEOMETRY Point 4326)	UNIQUE (`acknowledgement_no`), SPATIAL INDEX GiST (`location`)	Ingests citizen incident reports from NCRP / 1930 helpline feeds.
`alerts`	`id` (UUID), `target_account_id`, `risk_score` (FLOAT), `graph_score`, `geo_score`, `target_terminal_id`,	FOREIGN KEY to `accounts.id`, INDEX	Stores fused AI threat scores and structured SHAP

Table Name	Key Attributes & Data Types	Constraints & Indexes	Operational Purpose
	`explanation` (JSONB), `status` (VARCHAR)	(`risk_score` DESC, `status`)	feature attributions.
`patrol_units`	`id` (UUID), `callsign` (VARCHAR), `unit_type` (PCR_VAN/BEAT_MARSHAL), `current_location` (GEOMETRY Point 4326), `status`, `city`	SPATIAL INDEX GiST (`current_location`), INDEX (`city`, `status`)	Live location and dispatch state of mobile police field assets.
`audit_logs`	`id` (UUID), `officer_badge_id`, `action`, `target_id`, `details` (JSONB), `hash_signature` (VARCHAR 64 SHA-256), `timestamp`	INDEX (`officer_badge_id`), INDEX (`target_id`)	Cryptographic, tamper-evident record of all interdiction actions.
`restitution_orders`	`id` (UUID), `restitution_reference`, `victim_account_number`, `frozen_account_number`, `cfcfrms_lien_reference`, `amount_restituted`, `court_order_number`	UNIQUE (`restitution_reference`), FOREIGN KEY (`complaint_id`)	Governs Section 457 Cr.P.C. Magisterial release orders & reverse settlement.

2.3 Synthetic Fraud Benchmark Topology Modeling

Due to banking secrecy laws and the DPDP Act 2023, live transaction data cannot be exported. RakshaNet implements a calibrated synthetic transaction generator embedding three mathematical laundering topologies into background civilian traffic:

- **Star Dispersal (Fan-Out):** 1 primary compromise node transfers funds across 15 to 25 spoke accounts in bursts below ₹50,000 to evade statutory AML transaction thresholds.
- **Multi-Hop Linear Chains:** \$A \to B \to C \to D\$ (3 to 5 hops across different banks) with inter-hop delays of 3 to 8 minutes, defeating single-bank velocity heuristics.
- **Convergent Aggregation (Fan-In):** 5 to 10 intermediary layering nodes pool funds into a single cash-out node immediately prior to physical withdrawal.

3. DUAL-BRANCH AI FRAUD PREDICTION PIPELINE

3.1 Branch 1: Topological Graph AI (Inductive GraphSAGE)

Mathematical Foundation: Unlike transductive graph algorithms (e.g., DeepWalk, Node2Vec) that require retraining the entire network when a new account appears, GraphSAGE learns localized aggregator functions that generalize inductively to previously unseen nodes.

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Neighborhood Aggregation:  $h_{N(v)}^{(k)} = \text{AGGREGATE}_k ( \{ h_u^{(k-1)} \}, \text{forall } u \text{ in } N(v) )$ 
Node Representation Update:  $h_v^{(k)} = \text{sigma} ( W^{(k)} * [ h_v^{(k-1)} || h_{N(v)}^{(k)} ] )$ 
```

Where $h_v^{(k)}$ is node v 's embedding at layer $k=2$, $N(v)$ is a uniformly sampled neighbor set, and $W^{(k)}$ is a trainable projection matrix. A 2-layer mean aggregator is deployed to satisfy strict sub-50ms inference constraints.

Topological Link Prediction Head:

To evaluate if an unflagged account u is acting as a laundering link to suspect node v , learned embeddings are fused with graph-theoretic metrics:

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Adamic-Adar Proximity:  $A(u, v) = \sum_{w \in N(u) \cap N(v)} [ 1 / \log |N(w)| ]$ 
Link Probability:  $P(\text{MuleHop}_{\{u,v\}}) = \text{sigma} ( z_u^T * W_{\text{pred}} * z_v + \beta * A(u, v) )$ 
```

3.2 Branch 2: Geospatial ATM Cashout Forensics (HDBSCAN + XGBoost)

While fund transfers happen in milliseconds, physical ATM cashouts require foot travel and multiple withdrawals due to per-card daily withdrawal caps (₹20,000–₹50,000). RakshaNet predicts physical withdrawal locations using a two-stage spatial model:

- Stage 1 (HDBSCAN Density Clustering):** Clusters ATM locations and historical crime reports using mutual reachability distance based on PostGIS spherical geometry (`ST_DistanceSphere``):

```
 $d_{\text{mreach-}k}(a, b) = \max ( \text{core}_k(a), \text{core}_k(b), d_{\text{geo}}(a, b) )$ 
```

- Stage 2 (XGBoost Threat Scoring):** Evaluates individual ATMs within identified clusters using an engineered spatial-temporal feature matrix:

Feature Name	Parameter Formulation	Threat Hypothesis
<code>`dist_to_highway`</code>	$d_{\text{hwy}} < 300\text{m}$	Proximity to major expressways enables rapid vehicular getaway across police borders.
<code>`booth_security_rating`</code>	Binary flag (0 = Guarded, 1 = Unmonitored)	Isolated kiosks with single CCTV and no physical guard are heavily favored by cash runners.
<code>`nocturnal_hours`</code>	$t \text{ in } [23:00, 05:00] \text{ IST}$	Syndicates exploit off-peak hours when branch staff and pedestrian footfall are minimal.

Feature Name	Parameter Formulation	Threat Hypothesis
`temporal_velocity_decay`	$V(t) = V_0 * e^{-\lambda * t}$	Threat probability decays exponentially as elapsed time from initial debit increases.

3.3 Multi-Modal Risk Fusion & Explainable AI (SHAP)

The Composite Risk Score is calculated as: $S_{\text{fused}} = 0.60 * S_{\text{graph}} + 0.40 * S_{\text{geo}}$. To prevent black-box opacity in judicial proceedings, TreeSHAP decomposes the fused score into local feature attributions:

```
Shapley Value:  $\phi_i(x) = \sum_{S \subseteq F \setminus \{i\}} \frac{|S|!(|F| - |S| - 1)!}{|F|!} * [f(S \cup \{i\}) - f(S)]$ 
```

The system converts ϕ_i attributions into court-admissible legal text:

Court-Admissible Legal Explanation:
"Account #86174411141 scored 96.8% (Critical Threat): Topological bridge between two confirmed mule syndicates (+0.37 SHAP), High-risk ATM cluster proximity (+0.30 SHAP), Abnormal IFSC dispersal velocity (+0.15 SHAP)."

4.1 Legal Admissibility Compliance Matrix

Statutory Act / Section	Legal Requirement	RakshaNet Technical Implementation
Section 91 Cr.P.C. / Section 69B IT Act	Law enforcement authority to summon records and order emergency preservation of electronic assets.	Authenticated police officers click "Dispatch Freeze Order", transmitting signed Section 91 JSON payloads to core banking gateways.
CFCFRMS Gateway Protocol (I4C)	Inter-bank lien retainment preventing fund dissipation across commercial banking nodes.	Issues verified Bank Lien References (`SBI-CFCFRMS-20260909-XXXX`) and records retained fund volume in real-time.
Section 63 BSA 2023 / Section 65B IEA	Mandatory digital certificate certifying integrity of electronic computer output without tampering.	Generates statutory digital certificates sealed with a Merkle root SHA-256 digest, recording device ID, timestamp, and officer credentials.
Section 457 Cr.P.C. / Section 503 BNSS 2023	Magisterial court authorization required before seized/frozen funds can be debited and restored to the victim.	Automates Section 457 court petition drafting and executes one-click reverse RTGS settlement directly crediting victim accounts.
DPDP Act 2023 (Section 17(1)(c))	Data protection exemptions for prevention, detection, and prosecution of cyber offences.	Full statutory exemption; implements privacy-by-design with masked account numbers (`XXXX-XXXX-1141`) in presentation views.

```
Signature = SHA256 ( BadgeID || Action || TargetID || Timestamp || JSON(Details) )
```

4.3 Section 457 Cr.P.C. Citizen Restitution Lifecycle

6. Citizen Restitution <-- 5. Reverse RTGS Settlement <-- 4. Magisterial Petition
100% Funds Credited Debits Mule, Credits Chief Metropolitan
Verified in Victim Acc Defrauded Victim Account Magistrate Sanctions
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5. TACTICAL COMMAND CENTER & FIELD WORKFLOWS

5.1 User Interface Architecture

The presentation layer is built as a Single Page Application (SPA) using **React 18**, **Tailwind CSS**, and **Vite**. The interface uses a dark-mode military/tactical glassmorphism design optimized for 24/7 Cyber Police Control Rooms:

- Overview & Telemetry Dashboard (`/`):** Visualizes national metrics (Funds at Risk, Retained Funds, Golden Window Countdown), an interactive Leaflet GIS dark map with glowing red threat isochrones, and the live priority threat alert queue.
- Tactical Multi-Hop Command Center (`/command`):** Features an interactive Cytoscape.js force-directed graph displaying the victim debit node, intermediary mule hops, and the destination cashout terminal with live node traversal.
- Explainable AI Drawer (`CaseDrawer.jsx`):** Houses the suspect KYC dossier, bank lien confirmation badges, and the interactive SHAP feature importance bar chart.

5.2 Core Interactive Modals & Operational Modules

Modal Component	Trigger Location	Technical Capabilities & User Experience
<code>`ScenarioModal.jsx`</code> ⚡ Simulate Live Attack	TopBar Primary Action	Injects calibrated real-world Indian cyber scams (Mumbai UPI, Delhi Digital Arrest, Bengaluru Task). Executes graph traversals, AI scoring, and WebSocket broadcast in < 800ms with tactical audio radar chimes.
<code>`PatrolDispatchModal.jsx`</code>	CaseDrawer & Map Markers	Calculates nearest police PCR vans and Beat Marshals using PostGIS spatial indexing. Generates official flash orders (`MUM-PCR-FLASH-...`) with calculated Haversine distance and live ETA.
<code>`LegalDossierModal.jsx`</code>	CaseDrawer Action Bar	Renders official Government of India / State Police Cyber Cell letterhead with statutory Section 91 preservation notice, multi-hop transaction table, SHAP explainability panel, and Section 63 BSA certificate. Supports 1-click clean PDF printing (`window.print()`).
<code>`RestitutionModal.jsx`</code>	CaseDrawer ("Sec 457")	Renders side-by-side debit/credit cards (Debited Mule vs Credited Victim). Allows officer to enter Chief Metropolitan Magistrate order numbers and execute court-sanctioned reverse RTGS settlements.
<code>`VictimTrackModal.jsx`</code>	TopBar ("Citizen Portal")	Provides public, sanitized transparency for citizens tracking NCRP acknowledgement numbers (`NCRP-2026-MUM-8921`) across 4 transparent stages without leaking confidential police locations.

5.3 Real-Time WebSocket Telemetry Gateway

RakshaNet implements bidirectional WebSocket communication via native FastAPI WebSockets and Redis pub/sub. When an attack is simulated or a high-risk transaction is flagged:

1. The AI scoring engine publishes an event payload to Redis channel `alerts:broadcast`.
2. The WebSocket manager dispatches the payload to all connected browser clients in under 50ms.
3. The React client plays a synthesized tactical audio chime (Web Audio API) and animates the pulsing map marker on the Leaflet map without requiring a full page refresh.

6. AUTOMATED VERIFICATION & PERFORMANCE BENCHMARKS

6.1 Comprehensive Verification Results (100% Pass Rate)

The entire RakshaNet codebase is validated using standalone asynchronous Python verification suites that execute real HTTP requests, database transactions, and cryptographic validations against the live running stack:

Verification Suite	Script File	Assertions Checked	Pass Rate	Key Features Validated
Phase 5 Security Suite	`scripts/verify_phase5.py`	56 Checks	56/56 (100%)	JWT Auth, Route Guards, Concurrency Locks, Archival, Error Sanitization, Response Headers.
Phase 6 Demo & Gateway	`scripts/verify_phase6.py`	42 Checks	42/42 (100%)	Live Scenario Injection (<800ms), Bank Lien Ref, CFCFRMS Gateway, SHAP Explanations.
Phase 7 Patrol Interdiction	`scripts/verify_phase7.py`	42 Checks	42/42 (100%)	PostGIS Spatial Queries, PCR Dispatch Orders, ETA Calculation, WebSocket Handshake.
Phase 8 Legal Evidence	`scripts/verify_phase8.py`	50 Checks	50/50 (100%)	Section 65B Digital Certificate, Merkle Root SHA-256 Digest, Syndicate Profiling.
Phase 9 Citizen Restitution	`scripts/verify_phase9.py`	47 Checks	47/47 (100%)	Section 457 Court Petition, Reverse RTGS Settlement, Public 4-Stage Victim Tracking.
TOTAL REGRESSION SUITE		237 Checks	237/237 (100%)	ZERO REGRESSIONS ACROSS ENTIRE PLATFORM

6.2 System Latency & Performance Benchmarks

Operation / Workflow Stage	Current Industry Benchmark	RakshaNet Measured Performance	Efficiency Gain
End-to-End Interception Latency	24 to 72 Hours (Manual inter-agency emails)	740 milliseconds (Automated AI pipeline)	> 100,000x Faster
Multi-Hop Graph Traversal (2-Hop)	Manual batch Excel joins (Minutes / Hours)	38 milliseconds (Neo4j native Cypher)	Real-time streaming capability
Nearest Patrol Van Spatial Search	Manual police radio broadcast (5–15 mins)	12 milliseconds (PostGIS `ST_DWithin`)	Instantaneous asset dispatch

Operation / Workflow Stage	Current Industry Benchmark	RakshaNet Measured Performance	Efficiency Gain
Evidence Dossier Compilation	3 to 7 Days (Manual forensic transcription)	1.8 seconds (Automated SHA-256 brief)	Immediate court admissibility
Frontend Production Bundle	N/A	Vite build in 2.81 seconds (0 errors)	Production deployment ready

6.3 Conclusion & Deployment Readiness

RakshaNet represents a complete, mathematically rigorous, and legally compliant cyber-financial fraud interdiction platform. By bridging the critical gap between machine learning inference, banking execution protocols, and statutory Indian criminal procedure, the platform successfully protects citizen wealth during the vital Golden Window.